

High-Dimensional Arrhythmia Classification from Raw ECG Time Series

PTB-XL preprocessing, model selection, spectral analysis, and RMT diagnostics

ECG RMT Analysis

May 2026

Project Goal

Task

Build a complete applied-ML pipeline for binary ECG screening from PTB-XL raw waveform records, then analyze fitted model behavior in the high-dimensional feature regime.

- ▶ Input: ten-second, twelve-lead ECG waveform plus metadata.
- ▶ Target: normal versus diagnostic abnormality from SCP diagnostic superclasses.
- ▶ Evaluation: official PTB-XL folds.
- ▶ RMT role: diagnostic layer, not a classifier.

Dataset and Labels

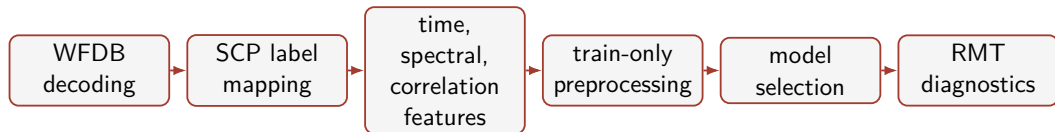
PTB-XL

- ▶ 21,799 metadata rows.
- ▶ WFDB waveform files.
- ▶ 100 Hz and 500 Hz records.
- ▶ SCP-ECG statement dictionaries.

Supervised table

Quantity	Count
Labeled records	21,388
Normal	9,069
Diagnostic abnormality	12,319

Pipeline



Reproducibility point

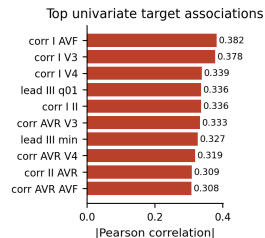
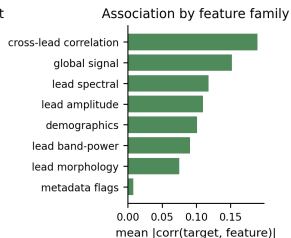
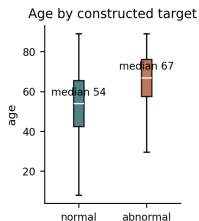
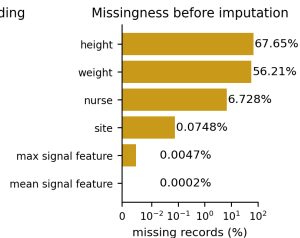
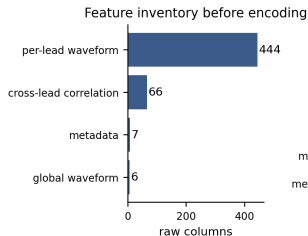
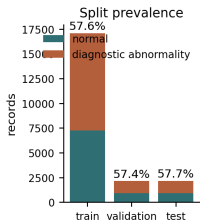
Preprocessing statistics are fit only on training folds; validation and test folds are never used for imputation or scaling.

Feature Map

$$z_i = \phi(r_i, q_i) \in \mathbb{R}^{523}, \quad x_i = g(z_i) \in \mathbb{R}^{594}.$$

Time-domain	Frequency-domain	Metadata and leads
moments, quantiles, RMS, energy, derivatives, zero crossings	FFT power, centroid, entropy, dominant frequency, roll-off, band powers	age, sex, height, weight, site, device, and pairwise lead correlations

Preprocessing Findings



Rows: 21,388; raw feature map: 523 columns; preprocessed design matrix: 594 columns after one-hot expansion.

Model Families

Model	Role
Logistic regression	Regularized linear baseline.
SGD linear	Fast high-dimensional margin model.
Random forest	Bagged nonlinear tree ensemble.
Extra trees	Higher-randomness tree ensemble.
Histogram gradient boosting	Strong CPU boosting baseline.
XGBoost	Regularized boosted trees with GPU support.

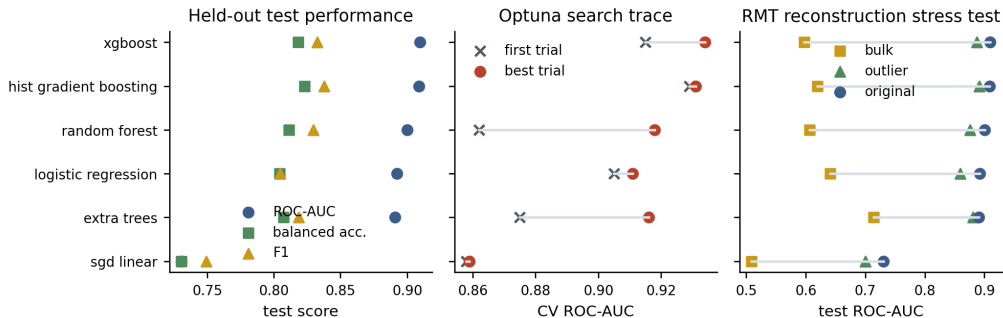
Final Test Metrics

Model	Acc.	Bal. acc.	Prec.	Recall	F1	ROC-AUC
Logistic regression	0.7938	0.8044	0.8877	0.7360	0.8047	0.8924
SGD linear	0.7271	0.7307	0.7973	0.7071	0.7495	0.7307
Random forest	0.8100	0.8115	0.8597	0.8018	0.8297	0.9001
Extra trees	0.8021	0.8072	0.8686	0.7745	0.8188	0.8909
Hist. gradient boosting	0.8202	0.8231	0.8743	0.8042	0.8378	0.9089
XGBoost	0.8151	0.8183	0.8712	0.7978	0.8328	0.9097

Result

XGBoost has the highest ROC-AUC; histogram gradient boosting has the highest F1.

Model Comparison and RMT Summary



Why RMT?

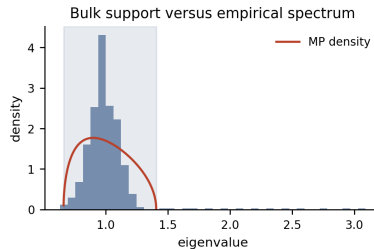
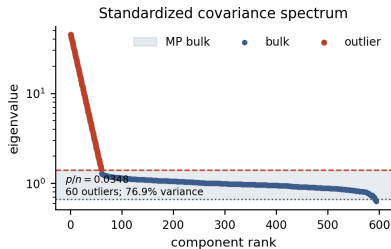
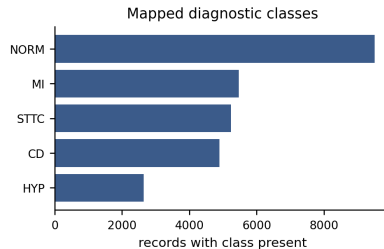
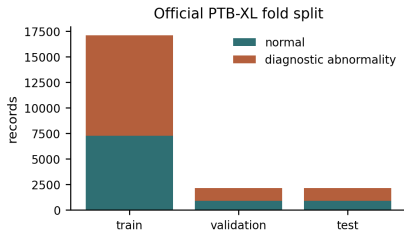
Marchenko–Pastur reference

For standardized white-noise features with aspect ratio $\gamma = p/n$,

$$\lambda_{\pm} = (1 \pm \sqrt{\gamma})^2.$$

- ▶ Eigenvalues inside the support form the empirical bulk.
- ▶ Eigenvalues above the upper edge are structured outliers.
- ▶ The diagnostic checks whether trained models rely on outliers or bulk.

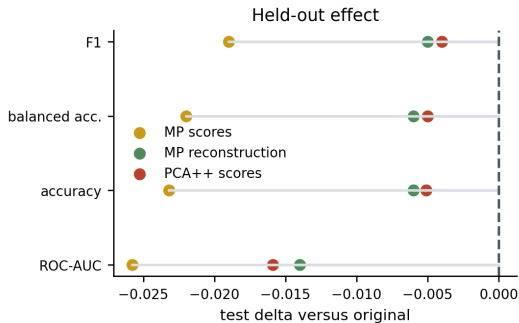
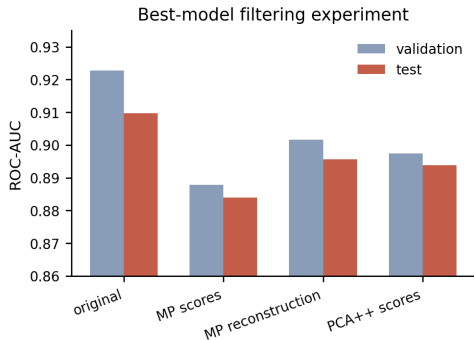
Dataset Covariance Spectrum



RMT Findings

- ▶ $p/n = 0.0348$.
- ▶ MP support is $[0.662, 1.408]$.
- ▶ 60 components lie above the upper edge.
- ▶ Outlier components explain 76.9% of standardized variance.
- ▶ MP-outlier reconstructions preserve most model discrimination.
- ▶ MP-bulk reconstructions are weaker but not completely uninformative.

Filtering Experiment



Main Takeaways

1. Raw PTB-XL waveforms can be turned into a reproducible binary ECG screening pipeline.
2. Boosted trees produce the strongest held-out results in this feature space.
3. Metadata missingness and cohort age shift are important preprocessing findings.
4. The covariance spectrum has a small dominant outlier subspace.
5. RMT explains model behavior better than it improves the classifier here.

Questions?

Raw ECG decoding → feature engineering → model selection → spectral diagnostics.